



Tyrrhenian Link - Commissioning Scenarios

SE GT GS CP STU

Scope of Work

Siemens Energy Grid Consulting

Revision 00

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Release

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Revision history

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1 Introduction

The following documents represents the technical offer from Siemens Energy (SE) Grid Consulting for the extension of the studies of the Tyrrhenian East Link for the assessment of two commissioning scenarios.

The mentioned studies will follow the same scope and methodology used for the evaluation in the Tyrrhenian East Link.

2 Scope of Work Included in the PO

2.1 Pos 50-a: Stability Modulation and Frequency Control Study

This study will be an extension of the stability study for Tyrrhenian East Link. The stability assessment will be performed for the two commissioning scenarios. In this sense, the study uses a different network configuration that deviates from the initial configuration in terms of number of lines and generation.

2.1.1 Objectives

The objectives of the study on "Stability, Modulation and Frequency Control" are:

- Evaluate the stability of the interconnected transmission system for system faults and other relevant contingencies. Stability in this context means the rotor angle stability of generators (synchronism), the stability of voltage and frequency in the power systems and the stable operation of the HVDC PLUS link.
- To investigate the need and, if so, to demonstrate the effectiveness of possible modulation functions (such as frequency control, power oscillation damping control, emergency power runbacks/runups) of the HVDC system
- To validate control parameters for stability functions implemented in the HVDC system.
- To avoid adverse interactions between frequency controllers (FC) and other primary response
- equipment.
- To investigate fundamental frequency interactions with other HVDC links, FACTS devices, renewable generation infeed nearby.

2.1.2 Methodology

Critical and realistic events are simulated in time domain for different representative load flow steady state conditions for the commissioning scenario of EAST LINK (1 Peak and 1 Offpeak scenario). Please note that the contractor will not add any additional load flow case other than the one's provided by the employer. While replacing the simplified HVDC model from the customer by the contractor's OEM HVDC model, the contractor will keep the same operating point (such as V, P and Q) at POI (point of interconnection) given by the network data from the employer.

The events are sensibly defined to:

- meet the requirements of the technical specification,
- demonstrate the stability of the system with HVDC PLUS in service,

- design the supplementary controls (such as frequency control, POD-control, AC voltage control, runbacks/runups if applicable in grid forming controls). A set of events agreed with the customer are analysed to reliably cover the objectives as listed above.
- If poorly damped system oscillation modes are not observed in the comprehensive network model provided by the employer, then a Power Oscillation Damping (POD) controller will not be designed or parameterized.

The simulated dynamic events are limited to 15 per converter station.

Control modes are limited to Q-mode and V-Mode

2.1.3 Input Data Required from Employer

Employer will provide to the Contractor:

- Updated AC network data (steady state and dynamic including power electronic devices near to the PCC) in Power Factory.

2.1.4 Deliverables

A study report including a description of the different tests, assumptions, operating points, results, and final conclusions and recommendations.

2.2Pos 60-a / 70-b: AC Network Reduction in Power Factory and implementation in PSCAD

2.2.1 Scope

Development of a PowerFactory DigSilent model, which is used for stability type studies, including modelling of the control required for phasor type digital models to investigate symmetric faults. All functions related to asymmetric functions are not part of model implementation. Very fast control loops acting in a transient range (time constants less than 1 ms) are not implemented in detail. The developed model to be delivered is for the use of dynamic power system simulations, in which the AC power system is typically represented by a positive sequence equivalent. DC Converters are represented by simplified equations and are not modelled to switching device level.

This study will be an extension of the Tyrrhenian East Link, but it requires to create additional AC equivalents for the two commissioning scenarios. The AC equivalents will follow the same methodology used in the Tyrrhenian East Link.

2.2.2 Deliverables

A RMS and PSCAD unencrypted models for the two commissioning scenarios reduced grids including the report with the description of the model, usage and benchmark with the original network.

3 Timeline

The following timeline applies to the proposed activities

		Project Schedule - Working Months																			
		0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Pos.	Description	M0					M1				M2				M3						
1	Pos 50-a Stability Modulation and Frequency Control Study																				
2	Pos 60-a AC Network Reduction Power Factory																				
3	Pos 70-b PSCAD Implementation of network reduction																				

M0

Contract award

M1

Completion Pos50-a

M2

Compection Pos60-a

M3

Completion Pos70-b

*Milestones to be completed at the end of the indicated month

Project meetings not shown here

4 Commercial Offer and Conditions

Any changes to this offer are subject to our written confirmation. The information, text and graphics provided in this document must not be used without prior written approval by Siemens Energy Grid Consulting for other purposes than the evaluation of this Technical Offer.

A brief portrait of the project including the customer name, the system/installation under consideration and keywords on the scope of work and on qualitative results may be used in Siemens Energy Grid Consulting's internal and external communication, unless the Client explicitly denies such use of project information.

The price information and reference to the applicable Commercial Conditions are included in the Commercial Offer which is provided as a separate document.

5 Quality Management

Our entire history has been shaped by innovation, by sophisticated high-tech products, and the unprecedented mastery and successful application of new methods and technologies. We have thus consistently been able to produce the very highest-quality products and services. This explains the great importance we attach to quality. However, the only way to achieve outstanding quality in our business is to permanently strive for it and design sustainable procedures to ensure it.

We have installed a comprehensive Quality Management Process, involving e.g. internal and external assessments of conformity with guidelines, laws as well as technical regulations and standards that are performed at regular intervals. External quality audits are conducted in response to special market, customer or official requirements. Internal audits are carried out once a year in all managing units in order to support and ensure that the mandatory quality elements are actually adhered to in practice. External audits are carried out in the cycles as specified in the respective guidelines and regulations.

The current certificate of the Quality and Environmental Management System issued by an independent assessor is shown in Figure 1.



CERTIFICATE

Management system as per

DIN EN ISO 9001 : 2015

DIN EN ISO 14001 : 2015

DIN ISO 45001 : 2018

The Certification Body TÜV NORD CERT GmbH hereby confirms as a result of the audit, assessment and certification decision according to ISO/IEC 17021-1:2015, that the organization

Siemens Energy Global GmbH & Co. KG

Division Transmission - Business Unit High Voltage Grids

Division Transmission - Business Segment Grid Stabilization

Division Transmission - Business Segment Service

Freyeslebenstraße 1

91058 Erlangen

Germany

with the locations according to the annex

operates a management system in accordance with the requirements of ISO 9001 : 2015, ISO 14001 : 2015 and ISO 45001 : 2018 and will be assessed for conformity within the 3 year term of validity of the certificate.

Scope

Division Transmission - Business Unit Transmission High Voltage Grids:

Development, design, project execution, production and sales of products, plants of high voltage direct current transmission as well as of maritime energy and transformer platforms

Division Transmission - Business Segment Grid Stabilization:

Development, design, project execution, and sales of products, plants of high voltage switchgear, high voltage reactive power compensation

Division Transmission - Business Segment Service:

Management of services activities: Field & maintenance services, repair & emergency services, modernization and retrofit service, spare part services, digital services, training services, support & consulting services, service programs incl. operation-management for products, systems and networks of power transmission & distribution

Certificate Registration No. 44 100 210961

Certificate Registration No. 44 104 210961

Certificate Registration No. 44 126 210961

Audit Report No. 3529 6183

Audit Report No. 3529 6184

Audit Report No. 3529 6185

Valid from 2021-07-29

Valid from 2021-07-29

Valid from 2021-07-29

Initial certification 2010

Initial certification 2010

Initial certification 2019

Valid until 2024-07-28

Valid until 2024-07-28

Valid until 2024-07-28

Certification Body
at TÜV NORD CERT GmbH

Essen, 2021-07-13

Validity can be verified at <https://www.tuev-nord.de/de/unternehmen/zertifizierung/zertifikatsdatenbank>.

TÜV NORD CERT GmbH

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45141 Essen

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Figure 1 Quality and Environmental Management System Main Certificate

6 Compliance

Siemens Energy shows zero tolerance toward corruption, violations of the principles of fair competition and other breaches of the law – and where these do occur, we take swift action. But compliance means much more than just adhering to laws and to the regulations described in the Siemens Energy Business Conduct Guidelines. Compliance forms the basis for all our decisions and activities, and it is the key to integrity when conducting business. Our premise is this: Only clean business is Siemens Energy business. This applies worldwide and at all levels of the organization. In addition to combating corruption and competition and export-control violations, the Compliance Department also protects our company against fraud and money laundering as well as safeguards personal data and human rights.

The compliance whistleblower hotline “Speak Up” at <https://www.bkms-system.net/Siemens-Energy/Speak-Up> gives all employees and external stakeholders a secure mechanism for the reporting of compliance violations.

7 Environmental Responsibility

Environmental protection is a corporate responsibility, social responsibility, and an important success factor for Siemens Energy. In all units of the company and in all countries in which we operate, it is our goal to protect the environment and conserve resources.

The Siemens Energy environmental programs are designed to conserve resources throughout the entire product life-cycle, reduce waste for disposal, and make our own business activities CO₂-neutral. The Siemens Energy Environmental Portfolio is our and our company's response to climate change, resource scarcity, and threats to the environment.

Our company meets the ecological demands of its partners by developing future-oriented and resource efficient solutions, products, and business models. Consistent and innovative environmental protection management is an integral part of our business processes and goes beyond legal requirements. We exert an influence on environmental impacts at an early stage in product and production planning, not only in the manufacturing phase, but also in the design, sales, utilization, service, and disposal phases. Climate protection plays a particularly important role for our company.